

O zdarzeniach $A, B \subset \Omega$. wiemy, że $P(A) = P(B) = \frac{1}{3}$ i $P(A \cap B) = \frac{1}{6}$

Oblicz: a) $P(A \cup B)$, b) $P(B - A)$

① DANE:

DO OBLICZENIA:

$$\begin{cases} P(A) = \frac{1}{3} \\ P(B) = \frac{1}{3} \\ P(A \cap B) = \frac{1}{6} \end{cases}$$

a) $P(A \cup B)$

b) $P(B - A)$

② a) $P(A \cup B) = P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cap \bar{B})$

$$P(A \cup B) = \frac{1}{3} + \frac{1}{3} - \frac{1}{6} = \frac{4}{6} - \frac{1}{6} = \frac{1}{2} \quad \text{odp: } \frac{1}{2}$$

③ b) $P(B - A) = P(B) - P(A \cap B)$

$$P(B - A) = \frac{1}{3} - \frac{1}{6} = \frac{2}{6} - \frac{1}{6} = \frac{1}{6} \quad \text{odp: } \frac{1}{6}$$

DODATEK:

możemy wykonać diagram Venna

